# SZ-7526TX
## Instruction Manual

Controlled cooling, always




**Touch Sensitive Temperature Controller**

## Introduction

The SZ-7526TX are the next generation of Subzero controllers. Their IP ratings are greatly improved and have an excellent iconic display. The touch feature whilst increasing reliability also gives a great user experience.

These controllers designed for refrigeration have several features specially designed for safety of compressors. They work on the system that:
**Cooling Mode:** The compressor output cuts off at set point and is restarted at a temperature of set point + differential.
**Heating Mode:** The compressor output cuts off at set point and is restarted at a temperature of set point - differential.
Their operation is very user friendly and is easily understood with the examples in the instructions below.

Various parameters help set up the instrument's functions for different applications.

**Features:**

- Range: -45.0°C to 100°C (when rS=0.1)
  -45°C to 100°C (when rS=1)
- Digital Input (Potential Free): AUX Digital Input for Compressor or Door Fault
- Compressor protection algorithm
- Auto/Man defrosting facility (Time based)
- TTL port compatible with RS485 Communication

## Cautions

**WIRING:** The probe and its corresponding wires should never be installed in a conduit next to control or power supply lines. The electrical wiring should be done as shown in the diagram. The power supply circuit should be connected to a protection switch. The terminals admit wires of upto 2.5sq mm.

**WARNING:** Improper wiring may cause irreparable damage and personal injury. Kindly ensure that wiring is done by qualified personnel only.

**Maintenance:** Cleaning: Clean the surface of the controller with a soft moist cloth. Do not use abrasive detergents, petrol, alcohol or solvents.

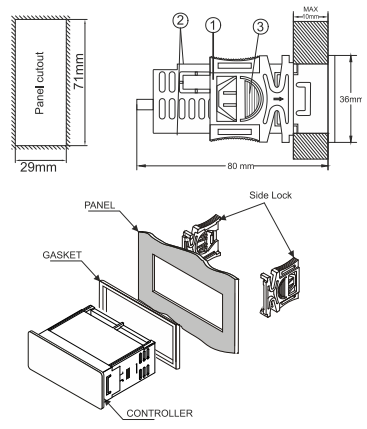
**Notice:** The information in this document is subject to change in order to improve reliability, design or function without prior notice and does not represent a commitment on the part of the company. In no event will the company be liable for direct, indirect, special, incidental or consequential damage arising out of the use or inability to use the product or documentation, even if advised of the possibility of such damages. No part of this manual may be reproduced or transmitted in any form or by any means without the prior written permission of the company.

**Installation:** Fixing and dimensions of panel models:
To fix the unit, slide the fastener (**1**) through the guides (**2**) as per the position shown in the figure. Move the fastener in the direction of the arrow, pressing tab (**3**) it permits to move the fastener in the opposite direction of the arrow.

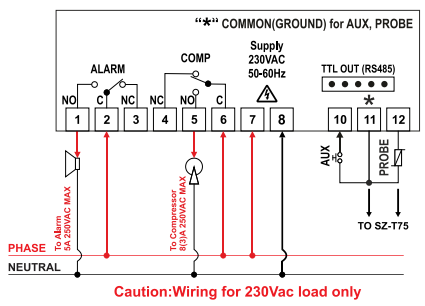
**Controller:** Controller should be installed in a place protected by vibration, water and corrosive gasses and where ambient temperature does not exceed the values specified in the technical data.

**Probe:** To give a correct reading, the probe must be installed in a place protected from thermal influences, which may affect the temperature to be controlled.

## Panel Cutout & Dimensions




## Suggested Wiring Diagram




Caution:Wiring for 230Vac load only

## Technical Data

| | |
|---|---|
| **Housing** | : Black ABS Plastic, Auto-extinguish |
| **Front Cover** | : Polycarbonate Plastic V0 Grade |
| **Dimensions** | : Frontal: 78 X 36mm, Depth: 80mm |
| **Panel Cutout** | : 29 X 71mm |
| **Mounting** | : Flush panel mounting with fasteners |
| **Protection** | : IP65 Front (with gasket) |
| **Connections** | : Screw terminal blocks.<br>< 2.5sq mm terminal only. |
| **Display** | : 2 X 17mm 7 segment display<br>1 X 11mm 7 segment display<br>6 LEDs for Indication |
| **Data storage** | : Non-volatile EEPROM memory. |
| **Power input** | : 230 Vac ± 10%, 50-60Hz Standard.<br>Others on request. |
| **Relay output** | : Compressor relay 8(3)A, 250V AC<br>Alarm relay 5A, 250V AC |
| **Operating temp.** | : 0°C to 60°C (non-condensing) |
| **Operating humidity** | : 20% to 85% (non-condensing) |
| **Storage temp** | : -25.0°C to 60.0°C (non-condensing) |
| **Measuring Range** | : -45.0°C to 100.0°C (when rS=0.1)<br>-45°C to 100°C (when rS=1) |
| **Input** | : NTC Control Probe |
| **Digital Input (Potential Free)** | : AUX Digital Input for Compressor or Door Open Fault |
| **Resolution** | : +/- 1°C / 0.1°C (Selectable) |
| **Accuracy** | : +/- 1°C |
| **TTL Connectivity** | : Baud Rate (Settable)<br>Device ID (Settable)<br>Parity (Settable) |

## User Interface

| | |
|---|---|
| ^ **UP KEY** | Touch and hold for 1 seconds to **view high/ low temperature logging.**<br>Used to **increment parameter values in Set Mode and Program Mode** and **scroll through parameters.** |
| PRG v **DOWN / PROGRAM KEY** | Touch & hold for 2 seconds **to enter in Program Mode.**<br>Used to **decrement parameter values in Set Mode and Program Mode** and **scroll through parameters.** |
| RST **DEFROST / RESET KEY** | Touch & hold for 500 msec **to Mute Alarm & manual Reset of Fault.**<br>Touch & hold for 2 seconds **to start manual defrost and stop auto/manual defrost.** Also used to **exit from Program Mode/ Parameters read only Mode.** |
| SET **SET KEY** | Touch & hold for 2 seconds **to enter in Set Mode.**<br>To **set and save parameters values** in Set Mode and Program Mode.<br>In high/ low temperature logging Mode touch & hold for 2 seconds to **reset high / low Temperature logged values.** |
| ^ + PRG v | Touch & hold for 2 seconds **to enter in Parameters read only Mode.** |
| PRG v + SET | Touch & hold to **see control probe temperature** if default display is other than Pr1. |

**Note:** If controller is in Set Mode/Program Mode/Parameters read only Mode, in case of inactivity of keys, it will return to parameter view mode in 30 seconds and will return to Main screen in next 30 seconds.

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**Parameter List**:

### 1. Set Point

Function: To set compressor relay set point

Touch & hold SET key for 2 seconds.

Display will show set value. The set point value can now be modified by using the UP/DOWN key. After selecting the desired value, touch the set key and user can see "- -" which confirms that the set point has been stored in memory.

| rS = 0.1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| LS+1.0 | HS-1.0 | 0.0°C |

| rS = 1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| LS+1 | HS-1 | 0°C |

### 2. To Set Other Parameters

Touch & hold PRG key for 2 seconds

Display will flash "P I".
To select other parameters, use UP/DOWN keys.

### 3. P I Parameter

Function: To set Controller for Heating or Cooling

To change value use ^ PRG v keys. To set value touch SET key.

CoL(0) = Cooling Mode.
HEt(1) = Heating Mode.

| Min | Max | Fac. Set |
|---|---|---|
| CoL | HEt | CoL |

### 4. P2 Parameter

Function: To set maximum allowable high temperature limit

Example: If this parameter is set to 99.0°C and P2d is set to 1.0°C then if the temperature reaches or goes above 99.0°C, display will show Ht (High Temp.) indicating that the temperature has reached or gone above the value set in this parameter. Ht Fault will be cleared when temperature reaches or goes below (P2-P2d) means (99.0°C -1.0 °C = 98.0°C).

| rS = 0.1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| P3+1.0 | 99.0°C | 99.0°C |

| rS = 1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| P3+1 | 99°C | 99°C |

Ht
(Message on display)

### 5. P2d Parameter

Function: Differential to Reset High Temperature Fault

Example: As explained for P2 parameter.

| rS = 0.1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| 1.0°C | 10.0°C | 1.0°C |

| rS = 1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| 1°C | 10°C | 1°C |

### 6. P3 Parameter

Function: To set minimum allowable low temperature limit

Example: If this parameter is set to -40.0°C and P3d is set to 1.0°C then if the temperature reaches or goes below -40.0°C, display will show Lt (Low temp) indicating that the temperature has reached or gone below the value set in this parameter. Lt Fault will be cleared when temperature reaches or goes above (P3+P3d) means (-40.0°C+1.0°C = - 39.0°C).

| rS = 0.1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| -40.0°C | P2-1.0 | -40.0°C |

| rS = 1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| -40°C | P2-1 | -40°C |

Lt
(Message on display)

### 7. P3d Parameter

Function: Differential to Reset Low Temperature Fault

Example: As explained for P3 parameter.

| rS = 0.1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| 1.0°C | 10.0°C | 1.0°C |

| rS = 1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| 1°C | 10°C | 1°C |

### 8. HS Parameter

Function: To set Maximum set point limit

Once set at a particular value, this will not allow the set point to go above this value.

Example: Setting this parameter at 25.0°C will not allow the set point to go above 24.0°C (HS-1).

| rS = 0.1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| SP+1.0 | 99.0°C | 99.0°C |

| rS = 1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| SP+1 | 99°C | 99°C |

**SP= Set Point**

### 9. LS Parameter

Function: To set Minimum set point limit

Once set at a particular value, this will not allow the set point to go below this value.

Example: Setting this parameter at -10.0°C will not allow the set point to go below -9.0°C (LS+1).

| rS = 0.1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| -40.0°C | SP-1.0 | -40.0°C |

| rS = 1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| -40°C | SP-1 | -40°C |

**SP= Set Point**

### 10. P4 Parameter

Function: To set the differential for compressor relay ON condition

**In Cooling Mode:**
If The Set Point is Set at 10.0°C and Differential is set to 2.0°C, When the System reached 10.0°C, Compressor Relay will go OFF. Since the Differential is 2.0°C, Compressor Relay will Come ON (Restart) at 12.0°C (10.0°C + 2.0°C).

**In Heating Mode:**
If The Set Point is Set at 10.0°C and Differential is set to 2.0°C, When the System reached 10.0°C, Compressor Relay will go OFF. Since the Differential is 2.0°C, Compressor Relay will Come ON (Restart) at 8.0°C (10.0°C - 2.0°C).

| rS = 0.1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| 1.0°C | 20.0°C | 2.0°C |

| rS = 1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| 1°C | 20°C | 2°C |

### 11. P5 Parameter

Function: To set probe calibration

In time it may be possible that the display may be offset by a degree or so.
To compensate for this error, user may need to add or minus the degrees required to achieve the correct temperature.

Example: The temperature on the display is 28.0°C, whereas the actual temperature is 30.0°C. User will have to set the P5 parameter to 2, which means that once out of the programming mode, the temperature on display will be 30.0°C (28.0°C+ 2.0°C).

| rS = 0.1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| -10.0°C | 10.0°C | 0.0°C |

| rS = 1 | | |
|---|---|---|
| Min | Max | Fac. Set |
| -10°C | 10°C | 0°C |

### 12. P6 Parameter

Function: To set time delay between compressor relay restart

This parameter is used to protect the compressor from restarting in a short period of time.

Example: If this parameter is set at 3 minutes, the compressor relay goes OFF at the set point, it will not restart for a minimum of 3 minutes, even if the differential is achieved earlier. This parameter is good to protect the life of the compressor.

Flashing
Time delay in progress

| Min | Max | Fac. Set |
|---|---|---|
| 0 Min | 99 Min | 3 Min |

### 13. ot Parameter

Function: Minimum ON Time for Compressor relay

This parameter is used to protect the compressor so that there is enough time for oil to return back to the compressor. This delay starts once the compressor relay is ON.

Example: If this parameter is set at 1 minute and if the temperature is achieved before 1 minute, then the compressor relay will remain ON for minimum 1 minute, though set point is achieved.

| Min | Max | Fac. Set |
|---|---|---|
| 0 Min | 20 Min | 0 Min |

### 14. E I Parameter

Function: Compressor relay status in case of Probe Failure

When set to
oFF (0) = Compressor relay status is OFF.
on (1) = Compressor relay status is ON.
CYC (2) = Compressor relay performs a duty cycle for Cn for minutes OFF and Cy for minutes ON.

| Min | Max | Fac. Set |
|---|---|---|
| oFF | CYC | CYC |

### 15. Cn Parameter

Function: Compressor relay OFF time during probe fault

Example: If this parameter is set to 4 minutes, then compressor relay will be OFF for 4 minutes while performing the duty cycle.

| Min | Max | Fac. Set |
|---|---|---|
| 1 Min | 99 Min | 4 Min |

**Note:** If E I is not CYC, this parameter will not applicable.

### 16. CY Parameter

Function: Compressor relay ON time during probe fault

Example: If this parameter is set to 10 minutes, then compressor relay will be ON for 10 minutes while performing the duty cycle.

| Min | Max | Fac. Set |
|---|---|---|
| 1 Min | 99 Min | 10 Min |

**Note:** If E I is not CYC, this parameter will not applicable.

### 17. P7 Parameter

Function: This parameter is used to set the Duration for Defrost Cycle

This parameter is applicable for both Auto and Manual Defrost and specifies for how much time the Defrost will last.

Example: If this parameter is set to 30 Minutes and Defrost Frequency means PB parameter is set to 1 Hour, then 1 Hour after Power is applied to controller, defrost will take place for 30 minutes. This cycle will repeat after every 1 Hour.

ON
Defrost in progress

| Min | Max | Fac. Set |
|---|---|---|
| 0 Min | 99 Min | 30 Min |

**Note:** If P I is set to HEt, this parameter will not applicable.

### 18. PB Parameter

Function: This parameter is used to set the frequency for Defrost Cycle

Example: As explained for P7 Parameter.

| Min | Max | Fac. Set |
|---|---|---|
| 1 Hrs | 31 Hrs | 6 Hrs |

**Note:** If P I is set to HEt or P7 is 0, this parameter will not applicable.

### 19. AFO Parameter

Function: To activate or deactivate AUX Digital Input for Door or Compressor fault

When set to,

diS (0) = Digital input is disabled.
doP (1) = Door Open digital input is open connection will be a fault.
dCL (2) = Door Open digital input is close connection will be a fault.
CoP (3) = Compressor fault digital input is open connection will be a fault.

CCL (4) = Compressor fault digital input is close connection will be a fault.

| Min | Max | Fac. Set |
|---|---|---|
| dS | CCL | dS |

| 20. do1 / CF1<br>Parameter | Function: To set fault sensing delay for AUX Digital Input of Door or Compressor Fault |
|---|---|

If AFD is doP or dCL then this parameter name is do1 and if AFD is CoP or CCL then this parameter name is CF1.

Example: If CF1 = 5seconds and if AUX Digital Input for Door or Compressor Fault is present for 5 seconds then fault is detected.

| Min | Max | Fac. Set |
|---|---|---|
| 0 sec | 600 sec | 5 sec |

**Note:** If AFD is dS, this parameter will not applicable.

| 21. do2/CF2<br>Parameter | Function: To set Output status on AUX Digital Input for Compressor or Door Fault |
|---|---|

When set to,

Enb (0) = No action will be performed, only fault indication led will be ON at fault, with fault message.

EnA (1) = No action will be performed, only fault indication led will be ON, Alarm and its icon will be ON at fault with fault message.

CoF (2) = Compressor will be OFF, fault indication led will be ON, Alarm and its icon will be ON at fault with fault message.

| Min | Max | Fac. Set |
|---|---|---|
| Enb | CoF | Enb |

**Note:** If AFD is dS , this parameter will not applicable.

| 22. CF3<br>Parameter | Function: To set reset mode for Compressor output, on AUX Digital Input for Compressor Fault |
|---|---|

If AFD is CoP or CCL then only this parameter name is CF3.

AUt (0) = Auto Reset.

nAn (1) = Manual Reset after CF4 retrials in 1 hour.

Example: If this parameter set to " nAn" mode & CF4 is set to 5 then, AUX Digital Input fault for Compressor will be cleared after 5 retrials only after pressing reset key for 500 milliseconds.

If this parameter is set to "AUTO" mode then COMP. fault will be cleared automatically when it is healthy.

| Min | Max | Fac. Set |
|---|---|---|
| AUt | nAn | AUt |

**Note:** If AFD is dS / doP / dCL, this parameter will not applicable.
If CF2 is Enb / EnA, this parameter will not applicable.

| 23. CF4<br>Parameter | Function: No of retrials of compressor when Manual reset is selected |
|---|---|

If AFD is CoP or CCL then only this parameter name is CF4.

Example: As mentioned in CF3

| Min | Max | Fac. Set |
|---|---|---|
| 1 | 10 | 5 |

**Note:** If AFD is dS / doP / dCL, this parameter will not applicable.
If CF2 is Enb / EnA, this parameter will not applicable.
If CF3 is AUt, this parameter will not applicable.

| 24. do3<br>Parameter | Function: To set delay time for temperature updating at Door open digital input fault |
|---|---|

If AFD is doP or dCL then this parameter name is do3.

Example: This Parameter is set to 60sec, Control Probe Temperature is -18.0˚C & Door open condition occurs then Control Probe Temperature value -18.0˚C at Door open condition will be held for the 60sec even if Control Probe Temperature is rising.

After over the Temperature hold duration display temperature will be increased by 0.1˚C at every 1sec until it reaches current Control Probe Temperature.

| Min | Max | Fac. Set |
|---|---|---|
| 0 sec | 180 sec | 0 sec |

**Note:** If AFD is dS / CoP / CCL, this parameter will not applicable.

| 25. did<br>Parameter | Function: To set Device ID |
|---|---|

This parameter is used to set the Device ID for the controller.

This parameter will not change on factory set.
If it is set to dS (0), communication will be disabled.

| Min | Max | Fac. Set |
|---|---|---|
| dS | 250 | - |

| 26. bd<br>Parameter | Function: To change Baud Rate |
|---|---|

This parameter will not change on factory set.

Communication baud rate from 9600 to 57600bps.
1 = 9600bps
2 = 19200bps
3 = 38400bps

| Min | Max | Fac. Set |
|---|---|---|
| 1 | 3 | - |

**Note:** If did is dS, this parameter will not applicable.

| 27. PAr<br>Parameter | Function: Selection of Parity Bit |
|---|---|

This parameter will not change on factory set.

non (0): No Parity
Evn (1): Even Parity
odd (2): Odd Parity

| Min | Max | Fac. Set |
|---|---|---|
| non | odd | - |

**Note:** If did is dS, this parameter will not applicable.

| 28. Ccr<br>Parameter | Function: To reset Compressor run hours |
|---|---|

When set to,

no (0) = Don't clear compressor run hours.

YES (1) = Clear compressor1 run hours.

| Min | Max | Fac. Set |
|---|---|---|
| no | YES | no |

| 29. ddF<br>Parameter | Function: This parameter is used to select display while the Defrost Cycle is in progress |
|---|---|

When set to,

Pr1 (0) = Control Probe Temperature.

dF (1) = Defrost Label.

CHt (2) = Control Probe Temperature value at the time of Defrost Start (Hold Temperature)

| Min | Max | Fac. Set |
|---|---|---|
| Pr1 | CHt | Pr1 |

**Note:** If P1 is set to HEt or P7 is 0, this parameter will not applicable.

| 30. nd<br>Parameter | Function: Default (Normal) display |
|---|---|

Pr1 (0) = Control Probe Temperature.

SP (1) = Compressor Set Point.

| Min | Max | Fac. Set |
|---|---|---|
| Pr1 | SP | Pr1 |

| 31. dd<br>Parameter | Function: Delay the display of temperature |
|---|---|

Each value corresponds to 1 seconds, if the value is set to 1, it corresponds to 1 seconds, if it is set to 2, it corresponds to 2 seconds and so on.

Example: If this parameter is set to 1, temperature on the display will be updated after 1 seconds. The same value will be considered for calculation and logging.

Display delay parameter is applicable only when temperature is increasing (rising). When temperature is decreasing (falling) this parameter will not be applicable.
If this parameter is set to 0, this feature will be disabled.

| Min | Max | Fac. Set |
|---|---|---|
| 0 sec | 180 sec | 0 sec |

| 32. AL<br>Parameter | Function: To activate alarm |
|---|---|

When set to,

dS (0) = Alarm Relay is disable for all Faults.

AHL (1) = Alarm Relay will be active for Ht, Lt and other faults.

AHt (2) = Alarm Relay will be active for Ht and other faults except Lt.

ALt (3) = Alarm Relay will be active for Lt and other faults except Ht.

**Note:** Alarm Relay will activate for PFL/PFH & AUX Digital Input Fault, if this parameter is set to AHL, AHt,ALt.

| Min | Max | Fac. Set |
|---|---|---|
| dS | ALt | AHL |

| 33. Adt<br>Parameter | Function: This parameter is used to set the time delay at Power ON for Alarm Indication |
|---|---|

Example: If this parameter is set to 20 minutes, once the controller is powered ON, no fault indication will be activated for 20 minutes.
Alarm delay is used only for High Temperature and Low Temperature but not for Control Probe fail and other faults.

ON
Fault is present

| Min | Max | Fac. Set |
|---|---|---|
| 0 Min | 99 Min | 0 Min |

| 34. Add<br>Parameter | Function: This parameter is used to set the time delay to activate Alarm Relay at Door Open |
|---|---|

Example: If this parameter is set to 60sec, on door open fault alarm relay and its icon will be ON after 60sec on occurrence of door open fault.

| Min | Max | Fac. Set |
|---|---|---|
| 0 sec | 600 sec | 60 sec |

**Note:** If AL is dS, this parameter will not applicable.

| 35. rS<br>Parameter | Function: To set controller resolution |
|---|---|

When set to,

0.1 (0) = It will take all parameter in 0.1°C resolution.
1 (1) = It will take all parameter in 1°C resolution.

This parameter will not change on factory set.

| Min | Max | Fac. Set |
|---|---|---|
| 0.1 | 1 | - |

**Note:** Temperature and temperature related parameters will also change accordingly.

| 36. PA<br>Parameter | Function: To change Password |
|---|---|

User cannot enter into program mode & set mode, if correct password is not entered.

If the password is kept other then 0, user need to enter correct password to enter into set/program mode.

If password is 0, user can directly access set/program mode.

| Min | Max | Fac. Set |
|---|---|---|
| -999 | 999 | 0 |

| 37. LP<br>Parameter | Function: To lock keypad |
|---|---|

This parameter is used to lock the keypad so that tampering is not possible by by-standers.

UnL (0) = keypad unlocked.

LoC (1) = keypad locked.

When locked all parameters can only be viewed, but not modified.
**Note:** If LP parameter is set to LoC and if user tries to change any parameter value, "LP" will flash on the display.

LP
Flashing

| Min | Max | Fac. Set |
|---|---|---|
| UnL | LoC | UnL |

| 38. oEr<br>Parameter | Function: To Display the Software Version |
|---|---|

| 39. FS<br>Parameter | Function: To restore default settings of the controller. |
|---|---|

When set to YES all parameters are programmed to factory set values.
Useful to debug setting related Problems.

When set to,
no (0) = FS is disable.

YES (1) = FS as per default value.

| Min | Max | Fac. Set |
|---|---|---|
| no | YES | no |

| 40. EP<br>Parameter | Function: To end programming |
|---|---|
| To end programming press "SET" key | Once the key is pressed, the controller goes into the normal mode and displays the temperature. |

**Parameters read only Mode**

| 41. CrH<br>Parameter | Function: To view Compressor run Hours |
|---|---|
| It will display compressor run hours. | |

| 42. PBH<br>Parameter | Balanced defrost frequency hours |
|---|---|

PBH will show the remaining Defrost cycle frequency hours.
**Note:** If P1 is set to HEt or P7 is 0, this parameter will not applicable.

| 43. PBn<br>Parameter | Balanced defrost frequency minutes |
|---|---|

PBn will show the remaining Defrost cycle frequency minutes.

**Note:** If P1 is set to HEt or P7 is 0, this parameter will not applicable.

| 44. EP<br>Parameter | Function: To end programming |
|---|---|
| To end programming press "SET" key | Once the key is pressed, the controller goes into the normal mode. |

**LED Indications**

| LED | Status | Description | Parameter |
|---|---|---|---|
| ❄ | OFF | Compressor is OFF. | SP, P4 |
| | ON | Compressor is ON. | |
| ❄ | OFF | Compressor is not in Defrost. | - |
| | ON | Compressor is in Defrost. | P7 |
| 🕒 | OFF | Compressor is not in time delay. | - |
| | ON | Compressor is in ot time delay. | ot |
| | FLASHING | Compressor is in P6 time delay. | P6 |
| AUX | OFF | Digital input fault is absent. | AFD, CF1/ d01, CF2/ d02, CF4 |
| | ON | Digital input fault is present (Compressor / Door Open). | |
| | FLASHING | Compressor fault is in manual reset. | CF3 |
| 🔔 | OFF | If AL parameter is set to disable. If AL parameter is enable and no fault. | AL |
| | ON | Any fault is present: Control Probe Fail / Door Open / Compressor fault / Ht / Lt | |
| °C | ON | Control probe temperature and temperature related parameters in°C | - |

**Fault Messages**

| Fault Message | Description | Parameter |
|---|---|---|
| PFL | **Control Probe Temperature Fail Low** Control Probe Temperature circuit open or without probe, or temperature is < -45.0°C. | - |
| PFH | **Control Probe Temperature Fail High** Control Probe Temperature short circuit or temperature is > 100°C. | - |
| Lt | Low temperature fault | P3 |
| Ht | High temperature fault | P2 |
| do | Door open fault | AFD, d01, d02 |
| CF | Compressor fault | AFD, CF1, CF2, CF3, CF4 |

**Operating Messages**

| Message | Description | Parameter |
|---|---|---|
| LP | Keypad Lock | LP |

| | | |
|---|---|---|
| LL | Last low Temperature | - |
| LH | Last high Temperature | - |
| LrS | In temperature log function: When LL and LH values are cleared | - |
| PA | Password to Enter to get entry into Set Mode, Program Mode | PA |
| dF | Defrost in progress | P7 |
| rSt | Alarm mute and manual fault reset by RST key. | - |
| ArS | Alarm mute by RST key. | - |
| FrS | Manual fault reset by RST key. | - |
| ndF | No Action on Defrost | - |

**Password Function**

- **In Programme Mode:**
  Touch & hold "PRG" key for 2sec. Display will flash "P1" parameter if "PA" value is kept "0". If other than "0", then "PA" and "0" will flash. Use "^" and "PRG" keys to enter the password. On entering correct value, display will flash the first parameter "P1". User can scroll through parameters using "^" or "PRG" keys.
- **In Set Mode:**
  Touch & hold "SET" key for 2sec. Display will flash set point value if "PA" parameter value is kept to "0". If other than "0", then display will flash "PA" and "0". Use "^" or "PRG" to enter the password. On entering correct value, display will flash set point value. User can set desired value using "^"or "PRG" keys. To save the modified value use "**SET**" key.

**High and Low temperature logging function**

- **How to see the logged values:**
  LL: Last Low temperature
  LH: Last High temperature
  Touch and hold "^" key for 1 second display will flash "LL" and the corresponding temperature for 4 seconds. After this, display will flash "LH" and the corresponding temperature for 4 seconds and come out of Log mode and will display Control probe temperature.
- **How to reset the Logged values:**
  While the display is showing the logged values, if user touches & holds the "SET" key for 1sec, the logged values will be cleared and "LrS" will be displayed.
  **Note: High & Low temperature logs will be saved at every 10 mins.**

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**OTHER PRODUCTS**

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Cold Room Controller
Chiller Controller
Two Compressor Controller
Heating Controller
Humidity Controller
Pressure Controller

CASTLE®
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Globe Valves
Hand Valves
Flow Switches
Solenoid Valves

REV3 – 23.11.2023